PERCENTAGE CHANGE

If a value changes by a certain percentage, it is possible to calculate the new value. Conversely, when a value changes by a known amount, it is possible to work out the percent increase or decrease compared to the original.

Finding a new value from a % increase or decrease

To find how a 55% increase or decrease affects the value of 40, first work out 55% of 40. Then add to or subtract from the original to get the new value.

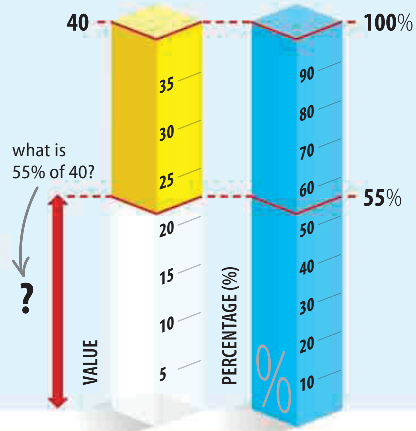
Known % ÷ 100 × Original value = % of total value THEN Original value + or − % of total value = New value

$$\frac{55}{100} \times 40 = 22 \qquad 40 \ {+ \atop \text{or} -} \ 22 = 62 \text{ or } 18$$

Divide the known % by 100 (55 ÷ 100 = 0.55).

Multiply the result by the original value (0.55 × 40 = 22).

Add the original value to 22 to find the % increase, or **subtract** 22 to find the % decrease.



Finding an increase in a value as a %

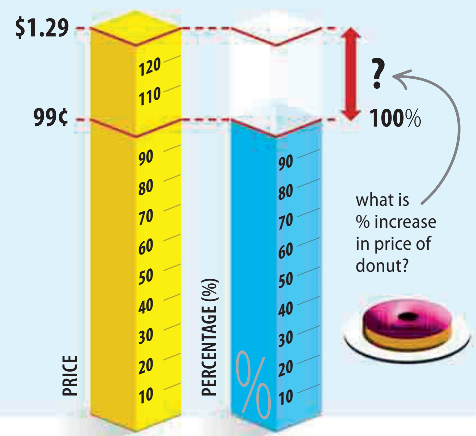
The price of a donut in the school cafeteria has risen 30 cents—from 99 cents last year to $1.29 this year. To find the increase as a percent, divide the increase in value (30) by the original value (99) and multiply by 100.

Increase in value ÷ Original value × 100 = Increase in value as %

$$\frac{30}{99} \times 100 = 30.3\% \text{ increase}$$

Divide the increase in value by the original value (30 ÷ 99 = 0.303).

Multiply the result by 100 to find the percentage (0.303 × 100 = 30.3), and round to 3 significant figures.



Finding a decrease in a value as a %

There was an audience of 245 at the school play last year, but this year only 209 attended—a decrease of 36. To find the decrease as a percent, divide the decrease in value (36) by the original value (245) and multiply by 100.

Decrease in value ÷ Original value × 100 = Decrease in value as %

$$\frac{36}{245} \times 100 = 14.7\% \text{ decrease}$$

Divide the decrease in value by the original value (36 ÷ 245 = 0.147).

Multiply the result by 100 to find the percentage (0.147 × 100 = 14.7), and round to 3 significant figures.